

Project proposal to be submitted for independent project dissertation for Information Technology

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Development of a comprehensive mobile application for export crop knowledge and trade facilitation



Introduction

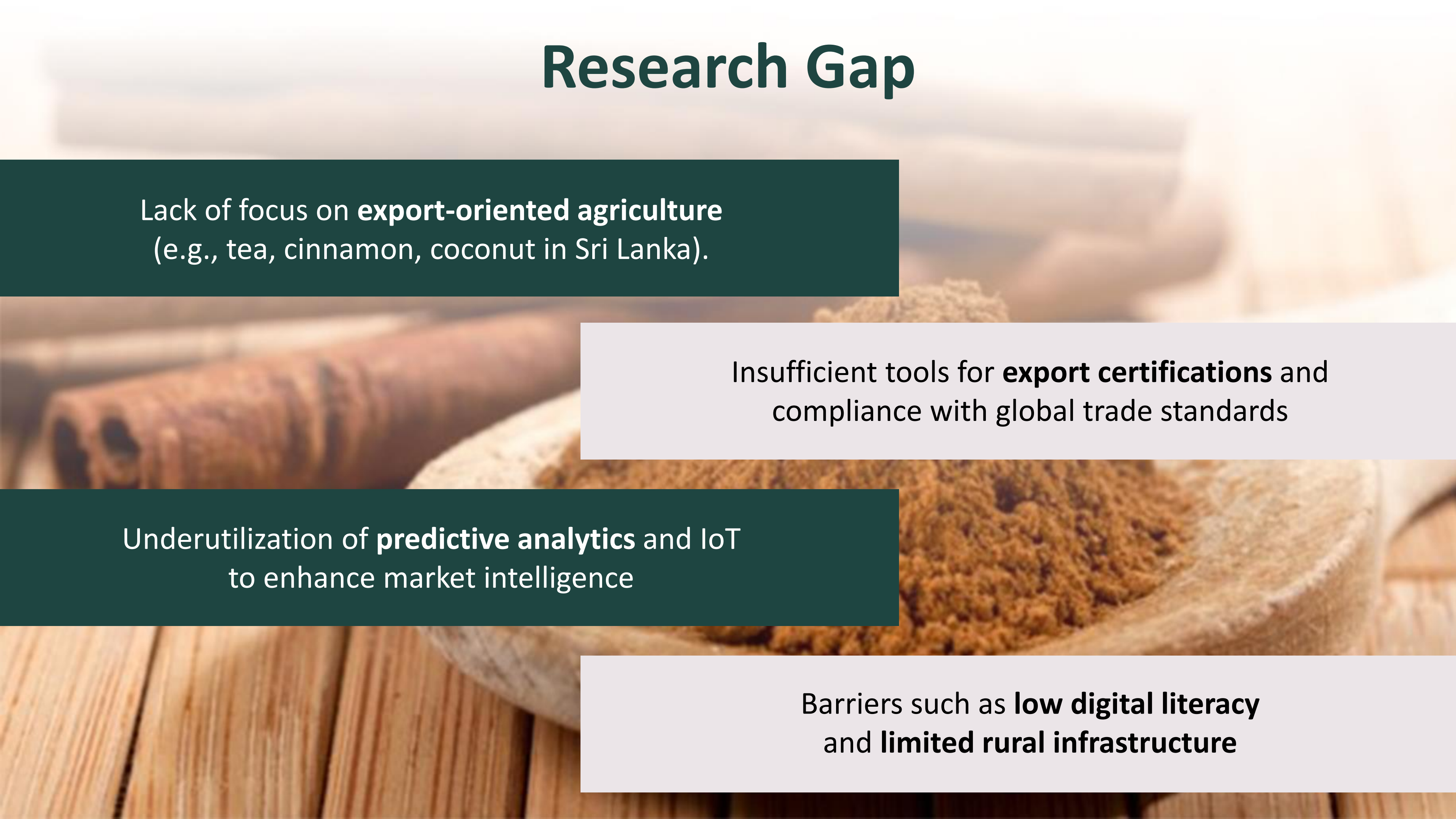
Sri Lanka plays a key role in the global agricultural market, especially through the export of tea, cinnamon, and coconut. However, farmers and exporters face challenges in accessing up-to-date information on market prices, trade regulations, and export opportunities. Existing resources, such as websites managed by the Sri Lanka Tea Board and the Export Development Board, are not easily accessible or user-friendly for small-scale farmers. This project aims to develop a mobile application that provides real-time updates, market insights, and trade information, helping to improve the efficiency and competitiveness of Sri Lanka's agricultural export sector.



Literature Review

Author(s)	Year	Field	Focus Area	Key Findings	Gap Identified
Shams et al.	2020	Mobile App Design	User-centric design in agriculture apps using human values theory	App alignment with user needs enhances adoption	Theoretical focus only; lacks export-oriented applications
Oteyo et al.	2021	IoT in Agriculture	Integration of IoT and AI for predictive farming solutions	Predictive analytics improves productivity and reduces waste	Lacks validation in real-world export agriculture contexts
Funmilayo & Ademola	2021	Agribusiness Development	IoT tools for agribusiness, improving productivity and local market access	IoT enhances operational efficiency and market connectivity	Focus limited to local markets without addressing global export challenges
Pachiyappan et al.	2022	Protected Cultivation	Economic benefits of controlled cultivation methods in high-value crops	Resource-efficient techniques improve profitability	Does not include export certifications or compliance for global trade
Burchi et al.	2018	ICT in Agriculture	Use of ICT-controlled greenhouses for optimized production	Technology enhances crop quality and minimizes resource wastage	Limited to controlled settings, excluding open-field agricultural applications

Research Gap



Lack of focus on **export-oriented agriculture** (e.g., tea, cinnamon, coconut in Sri Lanka).

Insufficient tools for **export certifications** and compliance with global trade standards

Underutilization of **predictive analytics** and IoT to enhance market intelligence

Barriers such as **low digital literacy** and **limited rural infrastructure**

Research problems



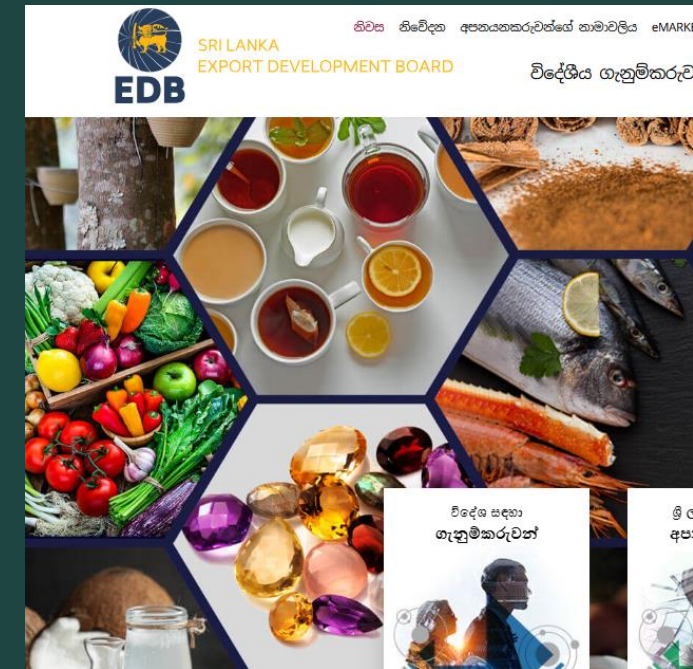
Limited Access to Reliable Information

Small-scale farmers and exporters in Sri Lanka struggle to access accurate, timely information about crop cultivation, market demand, and export regulations.



Insufficient Support for Small-Scale Exporters

Current platforms primarily cater to large-scale businesses, leaving small-scale farmers and exporters without the necessary tools to stay competitive in the market.



Non-User-Friendly Websites

The existing websites, such as those managed by the Sri Lanka Tea Board and Export Development Board, are not user-friendly, making it difficult for small-scale farmers and exporters to access relevant information



Missed Market Opportunities

Without up-to-date market information and access to real-time trade data, Sri Lankan exporters risk missing key opportunities for international trade.



Main Objective

To develop a comprehensive mobile application tailored for Sri Lanka's export crop sector (tea, cinnamon, and coconut), providing tools to enhance knowledge sharing, trade facilitation, and global market access for small-scale farmers and exporters.

Specific Objectives

To design a user-friendly platform that integrates real-time market data and production guidelines to support informed decision-making for farmers and exporters.

To provide export compliance tools, including guidelines on international certifications, quality standards, and best practices for tea, cinnamon, and coconut production.

To incorporate features for connecting farmers and exporters directly with international buyers, minimizing the role of intermediaries and improving profitability.

To promote sustainable agricultural practices by integrating eco-friendly farming guidelines and resource optimization techniques.

To address the digital literacy gap among small-scale farmers by designing a simple, accessible interface and offering localized training resources.

Significance of the Project

01

Revolutionizing Sri Lanka's Agricultural Export Sector

02

Improving Crop Quality and Sustainability

03

Marketplace Function to Boost Economic Growth

04

Promoting Digital Transformation in Agriculture



Methodology of the Study

01

Needs Assessment and Market Research Phase

- Qualitative Analysis
- Quantitative Analysis

02

App Design and Development Phase

- Prototype Development
- Agile Development Methodology
- User-Centered Design

03

Usability Testing and Evaluation Phase

- Usability Testing
- Data Collection
- Analysis



Problems, Limitations and Delimitations

- Connectivity Issues
- Target Audience
- Crop Selection
- Language Support
- Digital Literacy Challenges
- Scope of Information



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Thank You